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Java SE 27 & JDK 27

Class LinkedBlockingQueue<E>

java.lang.Object
  java.util.AbstractCollection<E>
    java.util.AbstractQueue<E>
      java.util.concurrent.LinkedBlockingQueue<E>

Type Parameters:

E - the type of elements held in this queue

All Implemented Interfaces:

Serializable, Iterable<E>, Collection<E>, BlockingQueue<E>, Queue<E>

```
public class LinkedBlockingQueue<E>
extends AbstractQueue<E>
implements BlockingQueue<E>, Serializable
```

An optionally-bounded blocking queue based on linked nodes. This queue orders elements FIFO (first-in-first-out). The *head* of the queue is that element that has been on the queue the longest time. The *tail* of the queue is that element that has been on the queue the shortest time. New elements are inserted at the tail of the queue, and the queue retrieval operations obtain elements at the head of the queue. Linked queues typically have higher throughput than array-based queues but less predictable performance in most concurrent applications.

The optional capacity bound constructor argument serves as a way to prevent excessive queue expansion. The capacity, if unspecified, is equal to `Integer.MAX_VALUE`. Linked nodes are dynamically created upon each insertion unless this would bring the queue above capacity.

This class and its iterator implement all of the *optional* methods of the `Collection` and `Iterator` interfaces.

This class is a member of the Java Collections Framework.

Since:

1.5

See Also:

Serialized Form

Constructor Summary

Constructors

Constructor	Description

`LinkedBlockingQueue()`	Creates a `LinkedBlockingQueue` with a capacity of `Integer.MAX_VALUE`.
`LinkedBlockingQueue(int capacity)`	Creates a `LinkedBlockingQueue` with the given (fixed) capacity.
`LinkedBlockingQueue(Collection<? extends E> c)`	Creates a `LinkedBlockingQueue` with a capacity of `Integer.MAX_VALUE`, initially containing the elements of the given collection, added in traversal order of the collection's iterator.

Method Summary

All Methods | Instance Methods

Modifier and Type	Method	Description
`void`	`clear()`	Atomically removes all of the elements from this queue.
`boolean`	`contains(Object o)`	Returns `true` if this queue contains the specified element.
`int`	`drainTo(Collection<? super E> c)`	Removes all available elements from this queue and adds them to the given collection.
`int`	`drainTo(Collection<? super E> c, int maxElements)`	Removes at most the given number of available elements from this queue and adds them to the given collection.
`void`	`forEach(Consumer<? super E> action)`	Performs the given action for each element of the `Iterable` until all elements have been processed or the action throws an exception.

Iterator<E>	**iterator**()	Returns an iterator over the elements in this queue in proper sequence.
boolean	**offer**(E e)	Inserts the specified element at the tail of this queue if it is possible to do so immediately without exceeding the queue's capacity, returning `true` upon success and `false` if this queue is full.
boolean	**offer**(E e, long timeout, TimeUnit unit)	Inserts the specified element at the tail of this queue, waiting if necessary up to the specified wait time for space to become available.
E	**peek**()	Retrieves, but does not remove, the head of this queue, or returns `null` if this queue is empty.
E	**poll**()	Retrieves and removes the head of this queue, or returns `null` if this queue is empty.
E	**poll**(long timeout, TimeUnit unit)	Retrieves and removes the head of this queue, waiting up to the specified wait time if necessary for an element to become available.
void	**put**(E e)	Inserts the specified element at the tail of this queue, waiting if necessary for space to become available.

int	**remainingCapacity**()	Returns the number of additional elements that this queue can ideally (in the absence of memory or resource constraints) accept without blocking.
boolean	**remove**(Object o)	Removes a single instance of the specified element from this queue, if it is present.
boolean	**removeAll**(Collection<?> c)	Removes all of this collection's elements that are also contained in the specified collection (optional operation).
boolean	**removeIf**(Predicate<? super E> filter)	Removes all of the elements of this collection that satisfy the given predicate (optional operation).
boolean	**retainAll**(Collection<?> c)	Retains only the elements in this collection that are contained in the specified collection (optional operation).
int	**size**()	Returns the number of elements in this queue.
Spliterator<E>	**spliterator**()	Returns a Spliterator over the elements in this queue.
E	**take**()	Retrieves and removes the head of this queue, waiting if necessary until an element becomes available.
Object[]	**toArray**()	

		Returns an array containing all of the elements in this queue, in proper sequence.
<T> T[]	**toArray**(T[] a)	Returns an array containing all of the elements in this queue, in proper sequence; the runtime type of the returned array is that of the specified array.

Methods declared in class AbstractQueue

add, addAll, element, remove

Methods declared in class AbstractCollection

containsAll, isEmpty, toString

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Methods declared in interface BlockingQueue

add

Methods declared in interface Collection

equals, hashCode, parallelStream, stream, toArray

Constructor Details

LinkedBlockingQueue

```
public LinkedBlockingQueue()
```

Creates a `LinkedBlockingQueue` with a capacity of `Integer.MAX_VALUE`.

LinkedBlockingQueue

```
public LinkedBlockingQueue(int capacity)
```

Creates a `LinkedBlockingQueue` with the given (fixed) capacity.

Parameters:

`capacity` - the capacity of this queue

Throws:

`IllegalArgumentException` - if `capacity` is not greater than zero

LinkedBlockingQueue

```
public LinkedBlockingQueue(Collection<? extends E> c)
```

Creates a `LinkedBlockingQueue` with a capacity of `Integer.MAX_VALUE`, initially containing the elements of the given collection, added in traversal order of the collection's iterator.

Parameters:

`c` - the collection of elements to initially contain

Throws:

`NullPointerException` - if the specified collection or any of its elements are null

Method Details

size

```
public int size()
```

Returns the number of elements in this queue.

Specified by:

`size` in interface `Collection<E>`

Returns:

the number of elements in this queue

remainingCapacity

```
public int remainingCapacity()
```

Returns the number of additional elements that this queue can ideally (in the absence of memory or resource constraints) accept without blocking. This is always equal to the initial capacity of this queue less the current `size` of this queue.

Note that you *cannot* always tell if an attempt to insert an element will succeed by inspecting `remainingCapacity` because it may be the case that another thread is about to insert or remove an element.

Specified by:

`remainingCapacity` in interface `BlockingQueue<E>`

Returns:

the remaining capacity

put

```
public void put(E e)
         throws InterruptedException
```

Inserts the specified element at the tail of this queue, waiting if necessary for space to become available.

Specified by:

`put` in interface `BlockingQueue<E>`

Parameters:

`e` - the element to add

Throws:

`InterruptedException` - if interrupted while waiting

`NullPointerException` - if the specified element is null

offer

```
public boolean offer(E e,
                     long timeout,
                     TimeUnit unit)
              throws InterruptedException
```

Inserts the specified element at the tail of this queue, waiting if necessary up to the specified wait time for space to become available.

Specified by:

`offer` in interface `BlockingQueue<E>`

Parameters:

`e` - the element to add

`timeout` - how long to wait before giving up, in units of `unit`

`unit` - a `TimeUnit` determining how to interpret the `timeout` parameter

Returns:

`true` if successful, or `false` if the specified waiting time elapses before space is available

Throws:

`InterruptedException` - if interrupted while waiting

`NullPointerException` - if the specified element is null

offer

```
public boolean offer(E e)
```

Inserts the specified element at the tail of this queue if it is possible to do so immediately without exceeding the queue's capacity, returning `true` upon success and `false` if this queue is full. When using a capacity-restricted queue, this method is generally preferable to method `add`, which can fail to insert an element only by throwing an exception.

Specified by:

`offer` in interface `BlockingQueue<E>`

Specified by:

`offer` in interface `Queue<E>`

Parameters:

`e` - the element to add

Returns:

`true` if the element was added to this queue, else `false`

Throws:

`NullPointerException` - if the specified element is null

take

```
public E take()
       throws InterruptedException
```

Description copied from interface: `BlockingQueue`

Retrieves and removes the head of this queue, waiting if necessary until an element becomes available.

Specified by:

`take` in interface `BlockingQueue<E>`

Returns:

the head of this queue

Throws:

`InterruptedException` - if interrupted while waiting

poll

```
public E poll(long timeout,
              TimeUnit unit)
       throws InterruptedException
```

Description copied from interface: `BlockingQueue`

Retrieves and removes the head of this queue, waiting up to the specified wait time if necessary for an element to become available.

Specified by:

`poll` in interface `BlockingQueue<E>`

Parameters:

`timeout` - how long to wait before giving up, in units of `unit`

`unit` - a `TimeUnit` determining how to interpret the `timeout` parameter

Returns:

the head of this queue, or `null` if the specified waiting time elapses before an element is available

Throws:

`InterruptedException` - if interrupted while waiting

poll

```
public E poll()
```

Description copied from interface: `Queue`

Retrieves and removes the head of this queue, or returns `null` if this queue is empty.

Specified by:

`poll` in interface `Queue<E>`

Returns:

the head of this queue, or `null` if this queue is empty

peek

```
public E peek()
```

Description copied from interface: `Queue`

Retrieves, but does not remove, the head of this queue, or returns `null` if this queue is empty.

Specified by:

`peek` in interface `Queue<E>`

Returns:

the head of this queue, or `null` if this queue is empty

remove

```
public boolean remove(Object o)
```

Removes a single instance of the specified element from this queue, if it is present. More formally, removes an element `e` such that `o.equals(e)`, if this queue contains one or more such elements. Returns `true` if this queue contained the specified element (or equivalently, if this queue changed as a result of the call).

Specified by:

`remove` in interface `BlockingQueue<E>`

Specified by:

`remove` in interface `Collection<E>`

Overrides:

`remove` in class `AbstractCollection<E>`

Parameters:

`o` - element to be removed from this queue, if present

Returns:

`true` if this queue changed as a result of the call

contains

```
public boolean contains(Object o)
```

Returns `true` if this queue contains the specified element. More formally, returns `true` if and only if this queue contains at least one element `e` such that `o.equals(e)`.

Specified by:

`contains` in interface `BlockingQueue<E>`

Specified by:

`contains` in interface `Collection<E>`

Overrides:

`contains` in class `AbstractCollection<E>`

Parameters:

`o` - object to be checked for containment in this queue

Returns:

`true` if this queue contains the specified element

toArray

```
public Object[] toArray()
```

Returns an array containing all of the elements in this queue, in proper sequence.

The returned array will be "safe" in that no references to it are maintained by this queue. (In other words, this method must allocate a new array). The caller is thus free to modify the returned array.

This method acts as bridge between array-based and collection-based APIs.

Specified by:

`toArray` in interface `Collection<E>`

Overrides:

`toArray` in class `AbstractCollection<E>`

Returns:

an array containing all of the elements in this queue

toArray

```
public <T> T[] toArray(T[] a)
```

Returns an array containing all of the elements in this queue, in proper sequence; the runtime type of the returned array is that of the specified array. If the queue fits in the specified array, it is returned therein. Otherwise, a new array is allocated with the runtime type of the specified array and the size of this queue.

If this queue fits in the specified array with room to spare (i.e., the array has more elements than this queue), the element in the array immediately following the end of the queue is set to `null`.

Like the `toArray()` method, this method acts as bridge between array-based and collection-based APIs. Further, this method allows precise control over the runtime type of the output array, and may, under certain circumstances, be used to save allocation costs.

Suppose x is a queue known to contain only strings. The following code can be used to dump the queue into a newly allocated array of `String`:

```
String[] y = x.toArray(new String[0]);
```

Note that `toArray(new Object[0])` is identical in function to `toArray()`.

Specified by:

toArray in interface Collection<E>

Overrides:

toArray in class AbstractCollection<E>

Type Parameters:

T - the component type of the array to contain the collection

Parameters:

a - the array into which the elements of the queue are to be stored, if it is big enough; otherwise, a new array of the same runtime type is allocated for this purpose

Returns:

an array containing all of the elements in this queue

Throws:

ArrayStoreException - if the runtime type of the specified array is not a supertype of the runtime type of every element in this queue

NullPointerException - if the specified array is null

clear

```
public void clear()
```

Atomically removes all of the elements from this queue. The queue will be empty after this call returns.

Specified by:

clear in interface Collection<E>

Overrides:

clear in class AbstractQueue<E>

drainTo

```
public int drainTo(Collection<? super E> c)
```

Description copied from interface: BlockingQueue

Removes all available elements from this queue and adds them to the given collection. This operation may be more efficient than repeatedly polling this queue. A failure encountered while attempting to add elements to collection `c` may result in elements being in neither, either or both collections when the associated exception is thrown. Attempts to drain a queue to itself result in `IllegalArgumentException`. Further, the behavior of this operation is undefined if the specified collection is modified while the operation is in progress.

Specified by:

drainTo in interface BlockingQueue<E>

Parameters:

c - the collection to transfer elements into

Returns:

the number of elements transferred

Throws:

UnsupportedOperationException - if addition of elements is not supported by the specified collection

ClassCastException - if the class of an element of this queue prevents it from being added to the specified collection

NullPointerException - if the specified collection is null

IllegalArgumentException - if the specified collection is this queue, or some property of an element of this queue prevents it from being added to the specified collection

drainTo

```
public int drainTo(Collection<? super E> c,
                   int maxElements)
```

Description copied from interface: BlockingQueue

Removes at most the given number of available elements from this queue and adds them to the given collection. A failure encountered while attempting to add elements to collection c may result in elements being in neither, either or both collections when the associated exception is thrown. Attempts to drain a queue to itself result in IllegalArgumentException. Further, the behavior of this operation is undefined if the specified collection is modified while the operation is in progress.

Specified by:

drainTo in interface BlockingQueue<E>

Parameters:

c - the collection to transfer elements into

maxElements - the maximum number of elements to transfer

Returns:

the number of elements transferred

Throws:

UnsupportedOperationException - if addition of elements is not supported by the specified collection

`ClassCastException` - if the class of an element of this queue prevents it from being added to the specified collection

`NullPointerException` - if the specified collection is null

`IllegalArgumentException` - if the specified collection is this queue, or some property of an element of this queue prevents it from being added to the specified collection

iterator

`public Iterator<E> iterator()`

Returns an iterator over the elements in this queue in proper sequence. The elements will be returned in order from first (head) to last (tail).

The returned iterator is *weakly consistent*.

Specified by:

`iterator` in interface `Collection<E>`

Specified by:

`iterator` in interface `Iterable<E>`

Specified by:

`iterator` in class `AbstractCollection<E>`

Returns:

an iterator over the elements in this queue in proper sequence

spliterator

`public Spliterator<E> spliterator()`

Returns a `Spliterator` over the elements in this queue.

The returned spliterator is *weakly consistent*.

The `Spliterator` reports `Spliterator.CONCURRENT`, `Spliterator.ORDERED`, and `Spliterator.NONNULL`.

Specified by:

`spliterator` in interface `Collection<E>`

Specified by:

`spliterator` in interface `Iterable<E>`

Implementation Note:

The `Spliterator` implements `trySplit` to permit limited parallelism.

Returns:

a `Spliterator` over the elements in this queue

Since:

1.8

forEach

```
public void forEach(Consumer<? super E> action)
```

Description copied from interface: `Iterable`

Performs the given action for each element of the `Iterable` until all elements have been processed or the action throws an exception. Actions are performed in the order of iteration, if that order is specified. Exceptions thrown by the action are relayed to the caller.

The behavior of this method is unspecified if the action performs side-effects that modify the underlying source of elements, unless an overriding class has specified a concurrent modification policy.

Specified by:

`forEach` in interface `Iterable<E>`

Parameters:

`action` - The action to be performed for each element

Throws:

`NullPointerException` - if the specified action is null

removeIf

```
public boolean removeIf(Predicate<? super E> filter)
```

Description copied from interface: `Collection`

Removes all of the elements of this collection that satisfy the given predicate (optional operation). Errors or runtime exceptions thrown during iteration or by the predicate are relayed to the caller.

Specified by:

`removeIf` in interface `Collection<E>`

Parameters:

`filter` - a predicate which returns `true` for elements to be removed

Returns:

`true` if any elements were removed

Throws:

`NullPointerException` - if the specified filter is null

removeAll

```
public boolean removeAll(Collection<?> c)
```

Description copied from class: AbstractCollection

Removes all of this collection's elements that are also contained in the specified collection (optional operation). After this call returns, this collection will contain no elements in common with the specified collection.

Specified by:

removeAll in interface Collection<E>

Overrides:

removeAll in class AbstractCollection<E>

Parameters:

c - collection containing elements to be removed from this collection

Returns:

true if this collection changed as a result of the call

Throws:

NullPointerException - if this collection contains one or more null elements and the specified collection does not support null elements (optional) or if the specified collection is null

See Also:

AbstractCollection.remove(Object), AbstractCollection.contains(Object)

retainAll

```
public boolean retainAll(Collection<?> c)
```

Description copied from class: AbstractCollection

Retains only the elements in this collection that are contained in the specified collection (optional operation). In other words, removes from this collection all of its elements that are not contained in the specified collection.

Specified by:

retainAll in interface Collection<E>

Overrides:

retainAll in class AbstractCollection<E>

Parameters:

c - collection containing elements to be retained in this collection

Returns:

true if this collection changed as a result of the call

Throws:

`NullPointerException` - if this collection contains one or more null elements and the specified collection does not permit null elements (optional) or if the specified collection is null

See Also:

`AbstractCollection.remove(Object)`, `AbstractCollection.contains(Object)`

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